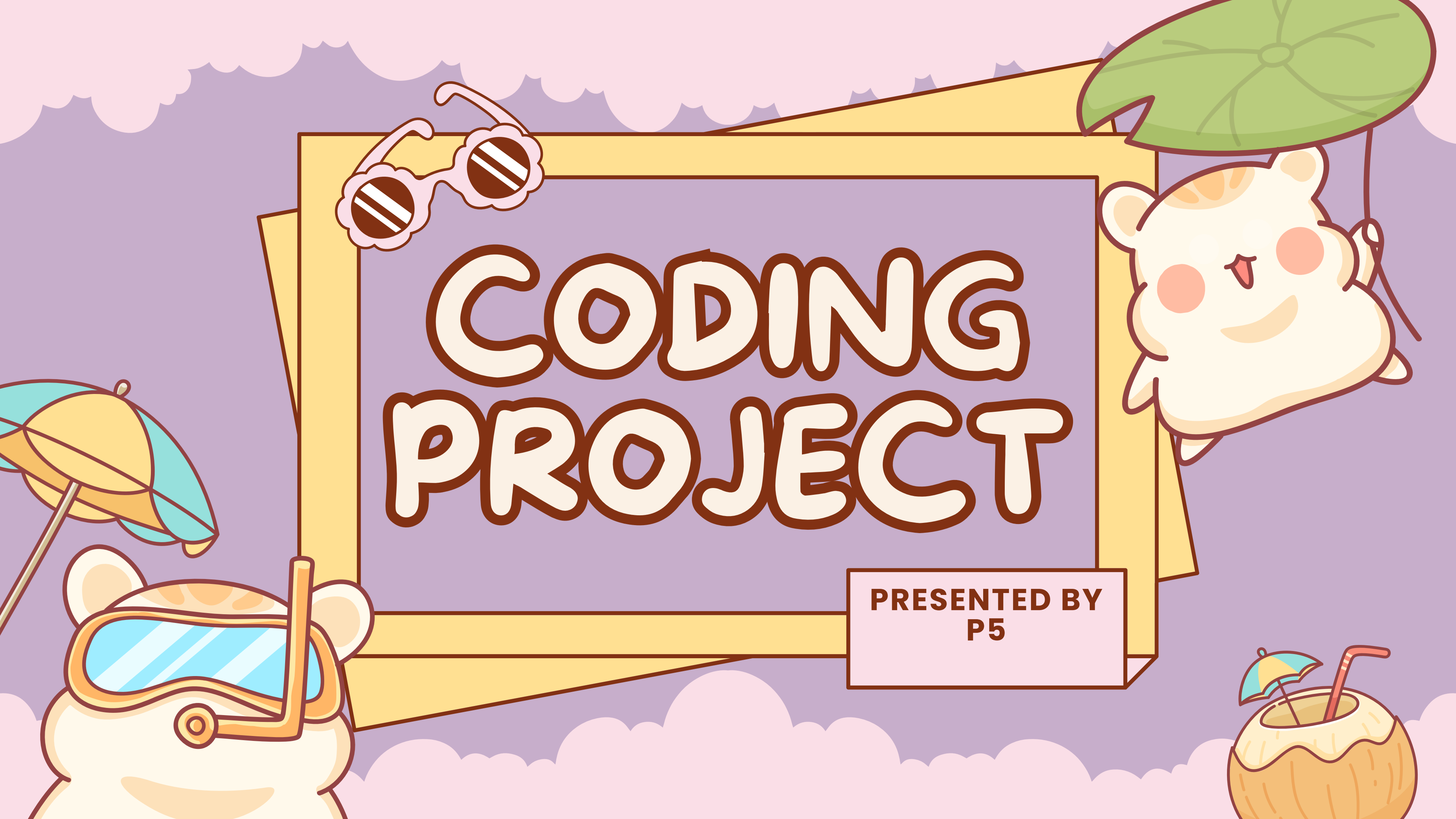




CODING PROJECT

PRESENTED BY
P5

MEMBER OF GROUP



HARSHEET



JOHN



PHEOBE



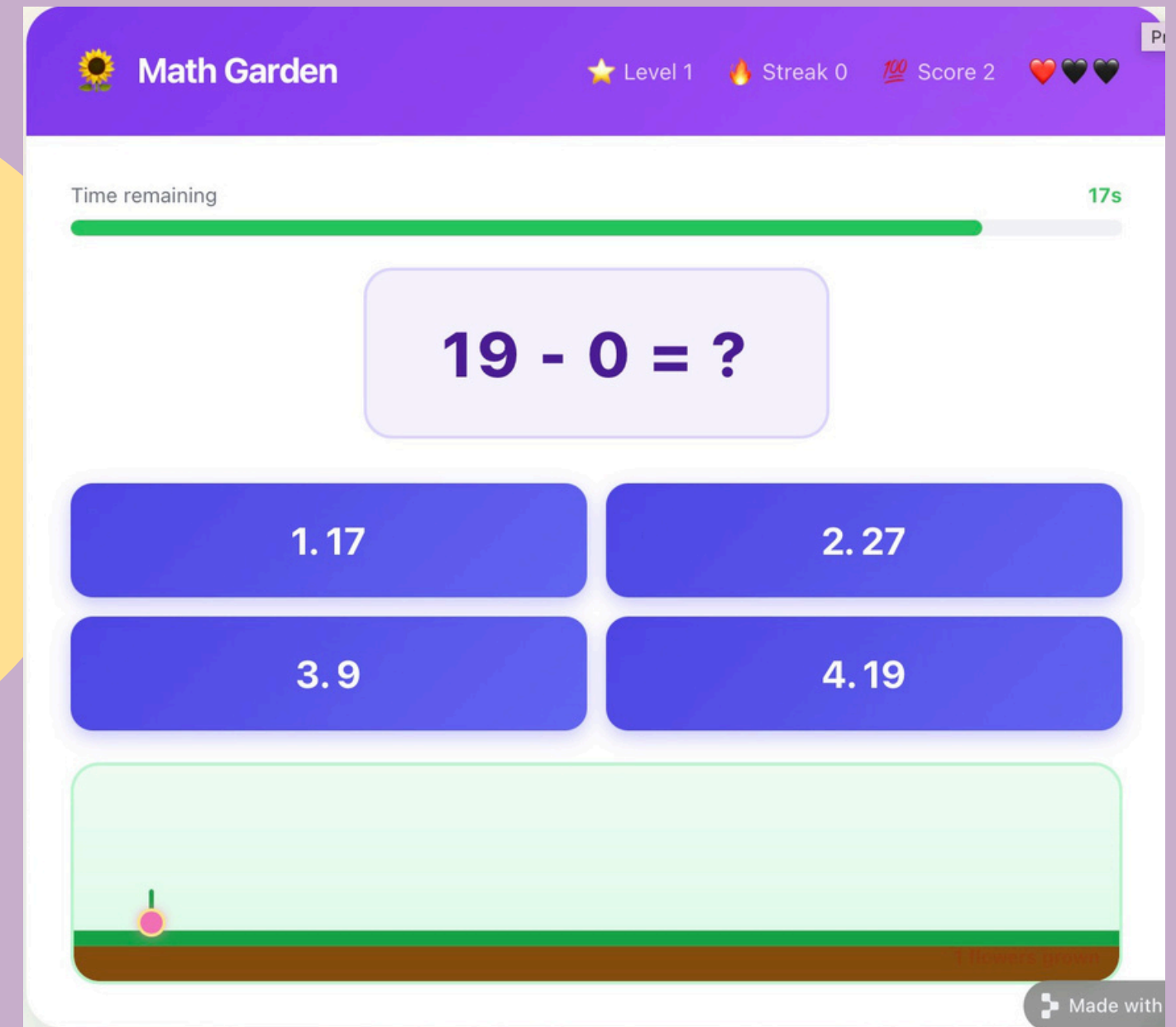
HANSINI



OUR MATH GARDEN

The entire gameplay revolves around solving basic arithmetic problems. The question generator progressively introduces addition, subtraction, multiplication, and division based on level.

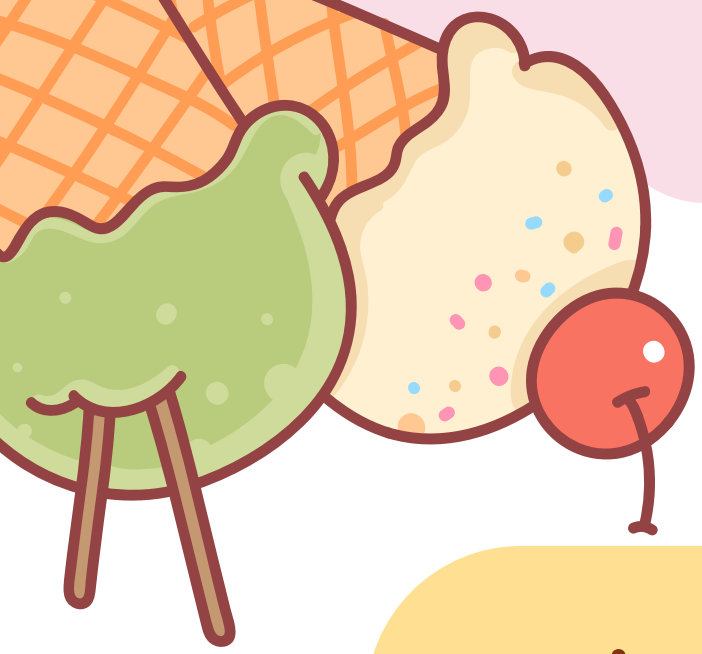
Since it is a 'Garden', every correct answer grows a cute little flower :).



TARGET AUDIENCE

Our target audience is children that are aged 8 and below to help stimulate their brain activity through these simple math problems





THE HIDDEN MAGIC

Doing mathematics worksheets in school is an experience that young kids do not always find pleasant, not necessarily because of the content at time but because it is a mundane activity.

This garden adds in the aspects of visual rewards, real progression, entertaining pressure and goal setting to an otherwise mundane question generator

The player is not just learning, they are growing something. The garden becomes a reflection of their success, making the experience feel more personal and rewarding for the player.



REWARD FACTOR

The game uses a rewarding progression system where every correct answer earns points, increases streaks, and grows a flower in the player's garden.

This provides immediate feedback and a visible sense of achievement. As players continue succeeding, they level up and unlock more challenging questions, encouraging motivation, persistence, and continued learning.

The audience is hit with small penalties if they get a question wrong, the reward system encourages positive reinforcement so that it creates an encouraging learning environment

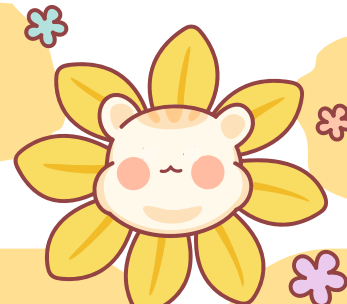




GARDEN QUALITIES

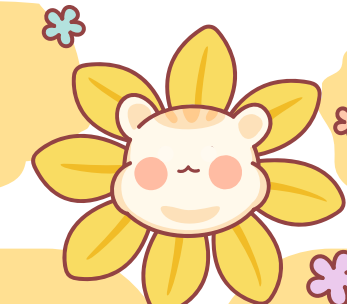
ENGAGEMENT & FUN FACTOR

Players earn points, build streaks, level up, and grow flowers in a garden for each correct answer. These gamification elements create motivation and reward progress. The timer also adds excitement.



VISUAL DESIGN

Uses colorful graphics, flowers, buttons, score displays, timer bars, and a garden visualization. The UI is simple but appealing for children.



USER GUIDANCE

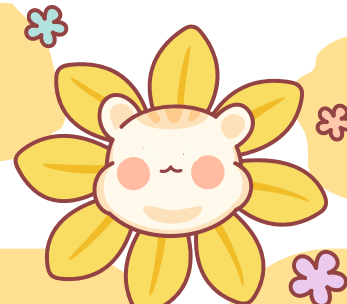
The question is clearly displayed, answer choices are shown as clickable buttons, and score/level/streak information is visible





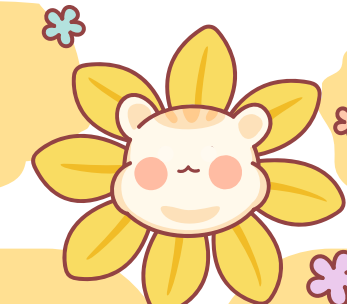
GARDEN QUALITIES

ORIGINALITY



Instead of simply answering math questions, players "grow a garden" through learning. The educational content is tied to a visual reward system, making the concept more memorable than a standard quiz.

ACCESSIBILITY AND USABILITY



There are large fonts, high contrast colors, and clickable buttons to help usability, alongside optional TTS.

CREATIVE USE OF TECHNOLOGY

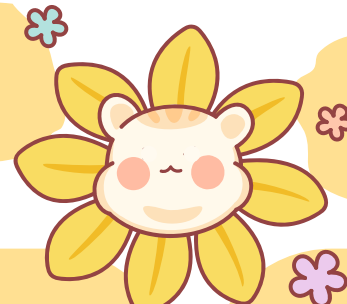
Combines Pygame graphics, procedural question generation, dynamic difficulty adjustment, timers, and optional text-to-speech. These technologies work together to create an educational experience.





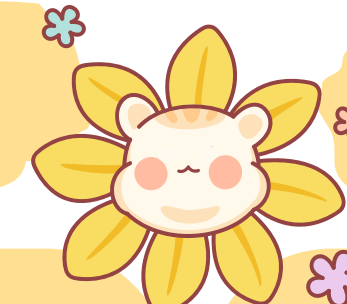
GARDEN QUALITIES

CHALLENGE INTERGRATION



The game includes multiple challenge systems: increasing difficulty levels, timed questions, scoring, streak bonuses, and penalties for wrong answers. This keeps gameplay progressively challenging.

MODULRITY AND FLEXIBILITY



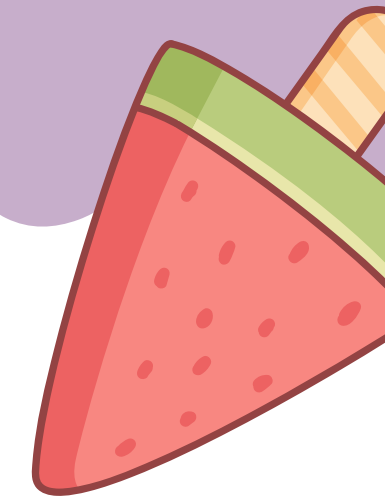
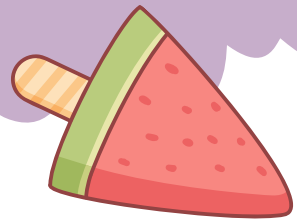
The code itself is well organised within separat, consistent of the 'Question generator', 'Garden', 'button', and 'game', which allows for the easy addition of new operations or garden elements without rewriting the whole code

SCALABILITY

The level system already supports expansion. Additional math topics, achievements, garden items, and player profiles can expand this demo version of the game into a bigger scale.



GAME DEMO!

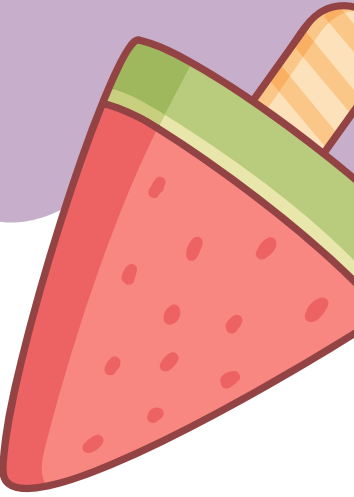
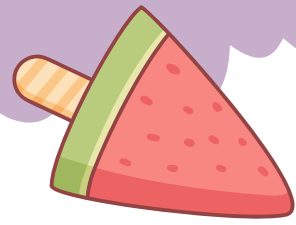


CONCLUSION

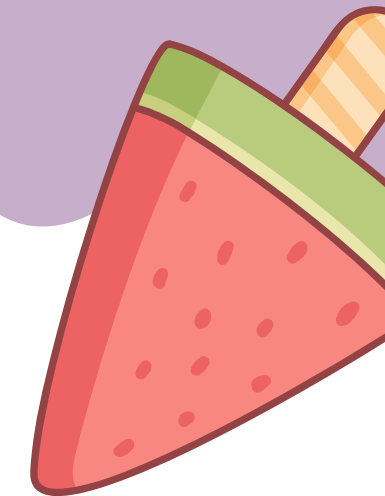
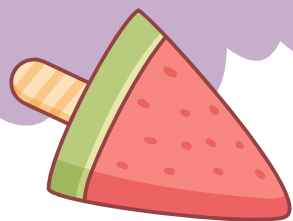
Math garden is more than just a question generator , more than just another application with ads or bad graphic's.

In heart it is a garden, a garden where one can grow.

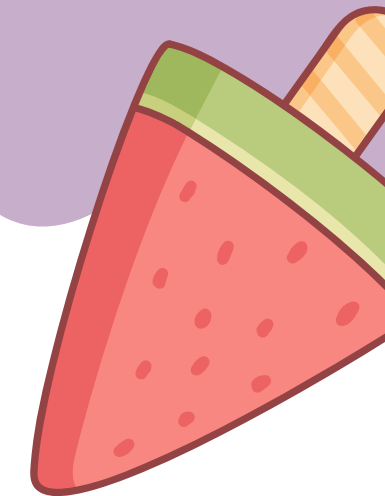
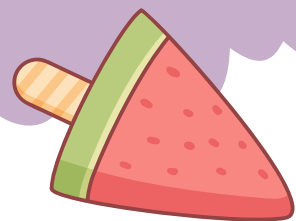
**Applications only offer a service, where one is the customer.
We want to offer a garden where children grow.**

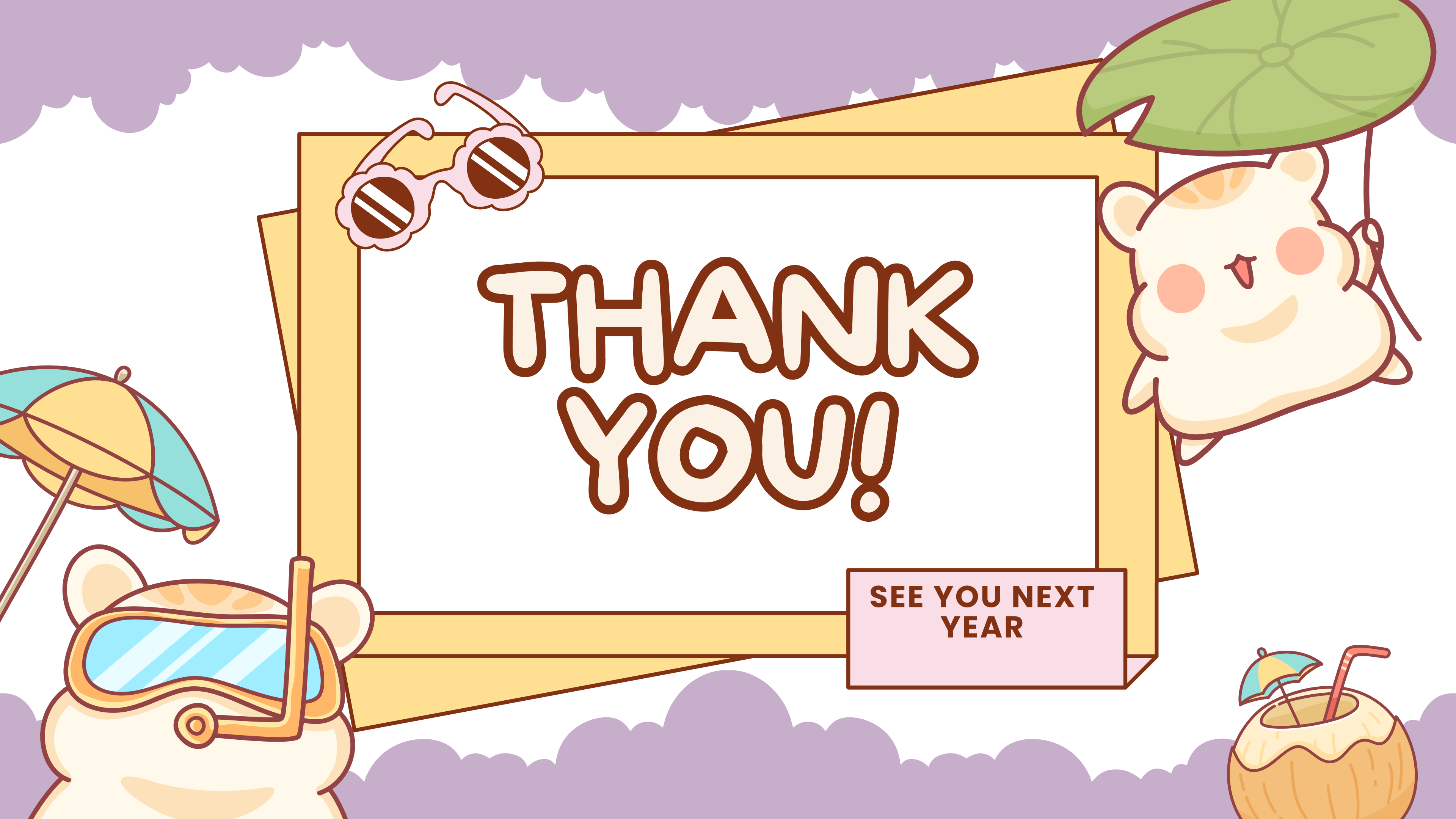


WE HAVE A
SLOGAN



KNOWLEDGE
HELPS THINGS
GROW





THANK
YOU!

SEE YOU NEXT
YEAR